Wireless Network design & Configuration

Wireless Network:

Frequency: 2.4GHz and 5GHz

Using 2.4GHz is the better option for its higher coverage to devices and wall penetration to wireless devices like printers. Using 5GHz is the better option for devices that need faster connections like laptops and phones.

What type of network: Infrastructure mode

Since business' require monitorization, security and scalability, having a single access point would make it the easiest to manage what is going in and out of the network.

Number of access points: 2

Having a two-story building with 15-20 users, each floor can be covered by one access point. These access points would be placed in the center of the ceilings which would help minimize the number of physical obstructions between the users and the access points

Type of antenna: Omnidirectional

Omnidirectional antennas equally signal in all directions. This is best when there are a lot of devices spread throughout the floor.

Wireless configuration:

SSID name: [business name]net-Secure, [business name]net-Guest.

The purpose of a guest network is to provide internet access to visitors, clients or people using their personal devices. This also isolates those devices from the corporate network which also protects any sensitive information from unauthorized access.

BSSID and SSID:

SSID is the actual human readable name of the wireless network. BSSID is a unique MAC address of the different access points handling the connection. It's how devices differentiate between different access points if they have the same SSID.

WPA3

WPA3 fixes certain bugs that fix the main security issues that WPA2 has. WPA3 is more protected against offline brute force attacks, it also has enhanced encryption for guest networks and individual data encryption for users on a shared network. WPA3 is more secure than WPA2.

Interference and Optimization:

2 Common sources of wireless interference:

Co-Channel interference from near by Wi-Fi networks and non-Wi-Fi interference from devices like microwaves, Bluetooth devices and walls.

How to reduce interference:

Using an automated channel to select clean channels, separate access points on non-overlapping channels, which prevents signal bleed and packet collisions.

Band Steering:

Band steering is a feature configured on dual-band access points that actively checks devices that can connect to 5GHz to connect to the 5GHz network.

Channel Bonding:

Channel bonding is connecting two or more channels to increase data throughput. It's useful during something that uses a lot of bandwidth, like video calls, or transferring large files.

Advanced Concepts:

Wireless Controller:

A device that manages, controls and secures multiple wireless access points across an organization.

Lightweight AP vs Autonomous AP:

Using a lightweight AP relies completely on a wireless controller for configuration, security and management. This is best for a medium to large AP where centralized control is required. An autonomous AP operates on its own and all AP's must be configured and individually monitored. This is best for small networks like a home office when a WLC is not needed.

Review:

Why are frequency ranges important in wireless networks?

Different frequencies have their own pros and cons, mainly between speed and coverage. They also help distribute traffic on the network across multiple channels to reduce bottlenecking.

What units measure signal strength and loss?

When Decibel-Milliwatts are measured, the closer they are 0, the stronger they are. The lower the negative number the weaker the signal is.

How does MIMO improve performance?

Multiple input multiple output uses multiple antennas on both the transmitter and receiver to send and receive multiple data streams at the same time. This increases the network capacity without needing extra power or bandwidth.

Why is channel bonding both helpful and risky?

Channel bonding can be good because it can increase speed and bandwidth, its better for resource heavy tasks like video calls or large file transfers. It also lowers the latency for single streams which reduces bottlenecking devices that need a quick response. Some drawbacks are increased congestion or interference because they use up available space quickly which leaves less non-overlapping channels for other devices. It also has a shorter range because of its higher frequency it works better close range.

Why is wireless security important?

As wireless signals are sent from device to device they can be intercepted by an attacker if they aren't encrypted properly. Wireless security also prevents people from unauthorized access to Wi-Fi which protects devices and internal files from being compromised.